

**HOW CONVAIR'S BIG B-36 BOMBER KEPT THE PEACE BY NEVER DROPPING A BOMB OR
SERVING IN COMBAT (UPDATED)**

THE BIRTH, LIFE AND DEATH OF THE "PEACEMAKER".

"FOUR BURNING & SIX TURNING"

555



HISTORYNET

7/26/2021

The Soviets have traditionally held primacy as aviation's gigantism specialists. Igor Sikorsky's Ilya Muromets, a vast stork of an airplane famously photographed with two crewmen strolling atop its fuselage, first flew in 1913. It was followed by a succession of Russian giants, including the [Tupolev ANT-20 Maxim Gorky](#), TB-3 bomber and Kalinin K-7. Even today, the world's heaviest airplane is the [Antonov An-225](#) Mriya, with a maximum takeoff weight of 705 tons.

But there was one interruption amid the steady stream of Soviet behemoths, and it flew at a time when the United States' archest enemy was that evil empire. When America needed a club with which to threaten the Russians, Convair produced the six-engine—eventually 10—B-36 long-range heavy bomber. It was the largest and heaviest piston-engine airplane ever to go into production.

The B-36 was nicknamed the Peacemaker, with a nod to the infamous Colt six-shooter. Some religious organizations objected, saying the only true Peacemaker was Jesus. That scared off the Air Force, which never called the airplane anything but B-36.

Many assume the B-36 was designed as a nuclear bomber, but the airplane's origins predate the possibility of any such mission. When first conceived during World War II, the big bomber was originally intended to reach Germany from the U.S. in the event that Britain fell. During its development, the B-36 was also touted for its potential ability to bomb Japan from Hawaii or Alaska.

When B-36s were slated to become nuclear-capable in the late 1940s, atomic weapons were controlled by the civilian Atomic Energy Commission and transferring them to the military was a cumbersome process.

Pre-strike bases had to be set up where a B-36 would land, refuel and pick up its bomb, the fissionable core of which presumably had been flown in by the AEC. Hardly a quick-reaction strike force.

These bases were all in remote northern locations, where technicians worked outdoors in bitter cold to assemble and load the bombs. Some B-36s were actually equipped with studded snow tires. In 1951 control of nuclear weapons was turned over to Strategic Air Command, but SAC's B-36s were still intended to stage and refuel through bases in Greenland, Canada and Alaska.



The XB-36 takes off on its first flight in August 1946. (U.S. Air Force)

The first XB-36 prototype had single-wheel main landing gear fitted with the largest aircraft tires ever manufactured—9 feet 2 inches in diameter. The configuration was chosen so it could easily retract into the wings. The extreme pressure of the tires' footprint, however, meant that only three runways in the world—at Carswell, Eglin and Fairfield (later Travis) Air Force bases—could bear the weight of the prototype. The single tires also meant that a blowout could wreck the bomber. Four-wheel bogies on each side of the airplane were quickly engineered to spread the load.

One challenging part of duty aboard an early B-36 was that, before landing, crewmen had to enter each wing—they were 7½ feet thick at the root—and clamber out to the gaping wheel well to confirm that the landing-gear down locks were correctly configured. "My first time, I was very scared," admitted ex-crewman Dick Graf. "I could look back and see that prop turning and knew that if I slipped I would be hamburger."

"If the gear didn't appear locked, we were supposed to hold on, stretch down between aircraft and oblivion and kick the [drag link] knuckle until a down lock was indicated," said retired Captain Reginald Beuttel Jr. "Talk about an exhilarating experience." Microswitches with cockpit indicators for the up and down locks were soon added.

The walk-in wings also made it possible to do some minor accessory-section work on the inboard engines. "Between the walkaround air bottle, the fresh-air rush, fuel and oil fumes, and the roar of the engines, doing the actual repair was simple, though it was always a memorable experience," recalled Staff Sgt. Bill Holding.

At some point early in the manufacturing process, Convair decided it needed to sweep the wings back a further three degrees to solve a center of gravity problem. The nacelles were already in place, with the engine centerlines established.

Rather than reengineer everything, the nacelles and mounts were left alone, so the B-36 flew with the propellers of its six pusher engines pointing three degrees inboard.



The B-36's remote-controlled turret guns were scaled up from the B-29's system. They proved unreliable and were later removed. (Joseph Scherschel/The LIFE Picture Collection via Getty Images)

The B-36 was originally intended to be a carpet bomber, spreading the largest number of iron bombs over the widest area. In a 1956 demonstration for congressmen and senators at Eglin AFB, a B-36 dropped a stick of 132 500-pound bombs in a line 2½ miles long—a conflagration that wouldn't be seen again until the advent of the B-52D "Big Bellies" over Vietnam. The 66,000-pound load was far from the B-36's max capacity—87,200 pounds for later models—and way beyond even a B-52's capability.

After a long, contentious and problematic development phase, the XB-36 made its first flight on August 8, 1946. Until the arrival of the B-36D in 1951, however, the airplane was not considered fit for combat. Its deeply buried Pratt & Whitney R-4360 pusher engines were difficult to cool, for the bitter-cold air at cruising altitude was too thin to get the job done. Carburetors would literally ice over, causing uncontained fuel spillage and fires. (One advantage of the pusher engines was that the crew could simply shut off the fuel feeding an engine fire and wait for it to blow out, with no damage to the wing's primary structure.) There were also constant propeller-vibration problems, and throughout the B-36's career some pilots would shut down the two inboard engines to prevent the hammering of propwash against the fuselage and the huge horizontal stabilizer.

"Keeping the B-36 in commission and battle-ready was a nightmare," recalled retired Staff Sgt. Manfred Wiest. "It would return from missions with pages of write-ups. It is questionable how serviceable it would have been in a combat situation."

The airplane's complex, radar-guided remote guns—a scaled-up B-29 system—were useless. The guns froze at altitude, their electronics interfered with every radio on the airplane and rarely could any gun fire a single belt of ammunition without a malfunction. Convair boasted that it was the most extensive defensive armament ever fitted to a warplane, though they failed to mention that it didn't work. Soon, B-36s were stripped of all but their tail guns.

Actually, the bomber's designers didn't intend the B-36 to defend itself with 20mm cannons. The plan was for the airplane to slowly labor to altitude—initially 40,000 feet, later 45,000 feet—where its thick wing continued to give it good handling qualities.

Fighter opposition would struggle to match its climb, if not its moderate 250-mph cruise speed, and once bomber and interceptor engaged, all the B-36 had to do was turn away from the fighter, which would be unable to follow the maneuver.

“We could evade them just by making a slight turn,” recalled former B-36 pilot David Flaming. “They could barely maneuver at those altitudes, and we could fly at 40, 45,000 feet pretty easily. You just alter your direction a little bit and they couldn’t compensate.”

When Chuck Yeager flew an F-86 in a test intercept against a B-36, he scored a few gun-camera hits but admitted that it was hard to hold a Sabre steady enough for accurate firing at 40,000 feet. Stripped and lightened late-model B-36s called Featherweights were able to cruise as high as 50,000 feet, and rumor has it that one special B-36 made it to 59,000.

Things would change by the mid-1950s with the arrival of the MiG-17 and then the MiG-19, and the first ground-to-air missiles. The B-36’s useful operational life was barely four years long.

The R-4360 Wasp Major was the B-36’s dedicated engine, an unreasonably heavy, 28-cylinder radial with four rows of cylinders, leading to the nickname Corncob. Despite its 3,000 horsepower—climbing to 3,800 hp in its most sophisticated, fuel-injected version, introduced in the B-36H—even six of them left the airplane underpowered. Convair wanted to pursue development of a more powerful engine, perhaps even a turboprop, but the Air Force’s budget wouldn’t allow it.



The B-36D was the first jet-equipped variant. (Universal Images Group via Getty Images)

So Convair added four General Electric J47 turbojets to the B-36D’s power-plant array. They were B-47 engines, affixed to B-36s still in their original two-engine Stratojet pods, and they burned avgas, not jet fuel. The jets were used to aid takeoff, but their real purpose was to augment dash speed over a Soviet target, which they did by 60 to 75 mph. Get in, get out and go home were the bywords for a bombing run, supposedly leaving fighters and anti-aircraft guns as little time as possible to identify and track an attacking B-36. Fire up the jets, two-block all 10 throttles and use the airplane’s substantial top speed—which, at 435 mph, was remarkable for a big bomber—to evade interception. Military planners seem to have ignored the fact that Soviet radar would have painted the lumbering bomber somewhere between the North Pole and the Barents Sea.

In 1966 the New York Central Railroad mounted a surplus B-36 dual-engine pod upside-down atop a Buddliner self-propelled diesel car and set an American rail speed record—183.68 mph—that stands to this day.

The Peacemaker lived its decade-long career largely as a PR tool. Even when the only B-36s flying were still in beta test, the Air Force sent them on low-altitude flights that would expose every large city in the country to their droning formations. Soviet spies were welcome to peek at the airplane on airshow ramps, since that was the point of the whole exercise: making sure Stalin knew that we had what Teddy Roosevelt called a “big stick.”

“Certainly, the Soviets quickly learned of the airplane’s many deficiencies as well. In the late 1940s, when just 40 B-36s were in squadron service, only five to eight were ever actually flyable. Still, the Soviets never came up with their own B-36 counterpart. The best they could do was the four-turbojet Myasishchev M-4 Bison strategic bomber, which had the range to reach the U.S. but not to return home. (When the Bison’s designer told Soviet Premier Nikita Khrushchev that the bomber could land in Mexico, Khrushchev responded: “Do you think Mexico is our mother-in-law? You think we can go calling any time we want?”)

And yes, B-36s were drones. During my teenage years in Yorktown Heights, N.Y., I would hear a B-36’s baritone as contrails etched the path of a tiny silver cross seven miles overhead. Nobody has ever authoritatively identified the source of the low-frequency, felt-as-well-as-heard growl, but the best guesses seem to be that it was caused by the hum of the near-supersonic prop tips, or that it was the sound of 18 prop blades cutting through the wash of air over the wings, or perhaps slicing through the engine exhausts.

Since no fighter had the range to escort a B-36 and aerial refueling was still under development, the B-36 was the focus of several parasite-fighter experiments. The best-known was the football-shaped McDonnell XF-85 Goblin, intended to be carried in one of the B-36’s bomb bays, to be launched from and retrieved by a hooklike trapeze. No B-36 ever carried a Goblin, but it was tested—unsuccessfully—using a B-29. Had the XF-85 ever gone into production, the best that could have been hoped was that MiG pilots would have died laughing.

Other experiments involved fighters towed along while attached to a B-36’s wingtips, an idea that foundered when wingtip turbulence caused two Republic F-84s and their B-29 towplane to crash in a ball of aluminum, killing all the participants. The most successful parasite utilized a full-size RF-84 snuggled up against a B-36’s belly. It was released not to wage combat but to do high-speed recon while its mother ship loitered on the safe side of the Soviet border. RB-36/RF-84 [FICON \(Fighter Conveyor\)](#) combos actually briefly became operational.

When the B-36 was designed, the engineers were well aware that it would be flying 30- to 40-hour missions, so they put extra effort into making crew accommodations as comfortable as possible: well-padded seats with armrests, carpeting, extensive soundproofing and insulation, built-in ashtrays, facilities for heating food, washbasins, beds, food-storage units and other luxuries never before seen on a bomber. In 1954, when the lightened B-36 Featherweights began to come online, these fripperies were the first to go. The Featherweights were developed to add enough range to eliminate the need for the Arctic staging bases. They also increased the B-36’s bombing altitude as well as its speed over the target.

The Air Force had asked Convair to engineer space for the 43,600-pound T-12 “bunker-buster” bomb, which was just over 30 feet long and 4½ feet in diameter. By doing so, they guaranteed that the B-36 would be able to carry the largest postwar atomic and hydrogen weapons. During its time in service, the Peacemaker was the only SAC bomber capable of carrying every bomb—conventional and nuclear—in the country’s arsenal.



A Republic RF-84 connects with its RB-36 mother ship during testing of the FICON (Fighter Conveyor) concept. (U.S. Air Force)

B-36s occasionally carried nuclear weapons, usually deactivated, and between 1952 and 1957 a group of them participated in a series of nuclear tests. Most of the experiments were in part directed toward assessing the effects of large ground or low-altitude explosions on the aircraft at altitude, the concern being that B-36s were too slow to escape serious shock-wave damage unless the nuclear weapon was parachute-retarded. In 1955, during Operation Teapot, B-36s performed three successful drops of low-yield 1.5- and 3.5-megaton weapons over a Nevada test range.



The Peacemaker's spacious bomb bays could accommodate the 21-ton Mark 17 thermonuclear bomb. (Margaret Bourke-White/The LIFE Picture Collection via Getty Images)

Nearly all of a B-36's fuselage was devoted to its two huge bomb bays, with a small, pressurized flight deck and crew compartment at the nose and a second one for gunners and spotters .

The pressurized portions of the fuselage are apparent because they are shiny aluminum, while the rest of the hull is dull magnesium, which doesn't take to the skin flexing of pressurization. The B-36 represents the largest use ever of magnesium in an airplane or spacecraft.

The Convair's long nose was flexible, and in turbulence it hunted from side to side enough to make the ride uncomfortable. "It could get rather violent at times," said Colonel Richard George. "I got calluses on my butt from the sideways motion." This also affected the forward guns, which could become misaligned with their sighting stations, and the accuracy of the bombsight. "They were never a lot of fun to fly," said ex-B-36 pilot Lt. Gen. James Edmundson. "It was like sitting on your front porch flying your house around."

One important variant, the RB-36 reconnaissance version, was largely intended to do follow-up photography of a just-struck target. The RB-36 served as a platform for a variety of cameras, the most sophisticated of which was the sole K-42 "Boston camera," so named because it was designed at Harvard and built by Boston University. The largest aerial camera ever made, with a 20-foot focal length, the Boston camera could photograph a golf ball from 45,000 feet. A photo displayed alongside the camera in the National Museum of the U.S. Air Force proves it, and an oblique shot of Manhattan taken from a distance of 72 miles shows individual New Yorkers strolling in Central Park.

In 1956 the first B-52s began replacing B-36s, and the Peacemakers began arriving at the Davis-Monthan AFB boneyard. They were immediately turned into aluminum and magnesium ingots. The last official B-36 mission was flown in February 1959, and SAC subsequently became an all-jet bomber force.

Of the 385 manufactured, few B-36s survive. One of the two original prototypes went to the Air Force museum, but they scrapped it. The late aviation collector Walter Soplata bought most of the fuselage and cockpit for \$760, and he stored it in his Ohio junkyard. It is now in the hands of a company in California that turns old airplanes into engraved metal "planetags" for enthusiasts.



Peacemakers parked at Davis-Monthan AFB await their final fate—scrapping and smelting into aluminum and magnesium ingots. (U.S. Air Force)

Four Peacemakers are still intact, though none will ever fly again. In Dayton, Ohio, the Air Force museum has a B-36J; the Pima Air & Space Museum in Tucson, Ariz., displays a B-36J; the Strategic Air Command & Aerospace Museum in Ashland, Neb., owns a B-36J; and a B-36H is parked at the Castle Air Museum in Atwater, Calif., east of San Francisco.

Was the B-36 an effective weapon, even though its operational career was short and pacifistic? Or was it yet another defense industry boondoggle, wreathed in controversy and corruption, born at the expense of the Navy's need for supercarriers while the bomber's performance was exaggerated?

Though the giant Convair was designed to be an offensive weapon, it turned out to be the biggest defense the U.S. offered to an enemy. Because the B-36 existed, the Soviet Union didn't risk starting a war in Europe during the opening decade of the Cold War. As imperfect as the B-36 was, it still was the first leg of what would become America's nuclear deterrent triad: very long-range strategic bombers, intercontinental missiles and nuclear-missile-equipped submarines. Legs two and three were yet to come, but the big Convair barred the door in the meantime.

Its message was a simple one—"don't you dare"—and the Peacemaker spoke effectively.

This feature originally appeared in the July 2021 issue of Aviation History.



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